# 7 ELECTRICITY

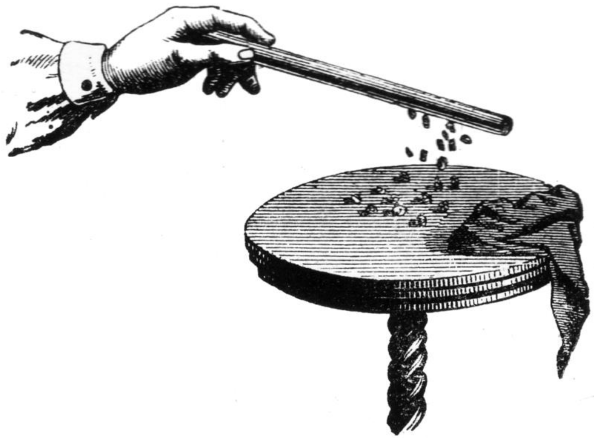
## CHAPTER OUTLINE




Vern J. Ostdiek
Donald J. Bord


CENGAGE

# 7.1 Electric Charge



- The word *electricity* comes from electron, which itself is based on the Greek word for amber.

- Amber is a fossil resin that attracts bits of thread, paper, hair, and other things after it has been rubbed with fur. You may have noticed that a comb can do the same after you run it through your hair.

# 7.1 Electric Charge

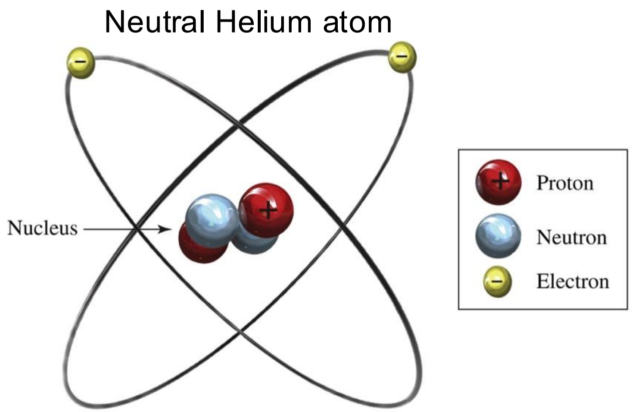
The electrical property was eventually traced to the atom and is called electric charge.

DEFINITION

**Electric Charge**: An inherent physical property of certain subatomic particles that is responsible for electrical and magnetic phenomena. Charge is represented by $q$, and the SI unit of measure is the *coulomb* (C).

Early on, we stated that there are three fundamental physical quantities: space, time, and matter. Until now, mass has been the only fundamental property of matter that we have used. Electric charge is another basic property of matter, but it is intrinsically possessed only by electrons, protons, and certain other subatomic or “elementary” particles. Unlike mass, there are two different (and opposite) types of electric charge, positive and negative.

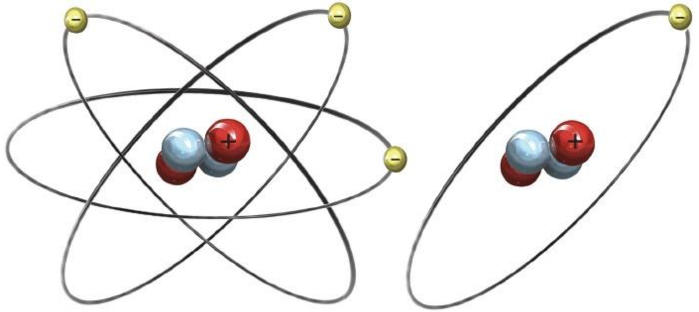
# 7.1 Electric Charge




- Every atom is composed of a nucleus surrounded by one or more electrons (have negative electric charge).
- The nucleus itself is composed of two types of particles: protons and neutrons.
  - Neutrons: have no net electric charge
  - Protons: have positive electric charge. The number of protons in the nucleus is the element's atomic number (Z).
- Normally, an atom will have the same number of electrons as it has protons, which means the atom as a whole is neutral.

# 7.1 Electric Charge

- A variety of physical and chemical interactions can cause an atom to *gain* one or more electrons or to *lose* one or more of its electrons. In these cases, the atom is said to be *ionized*.

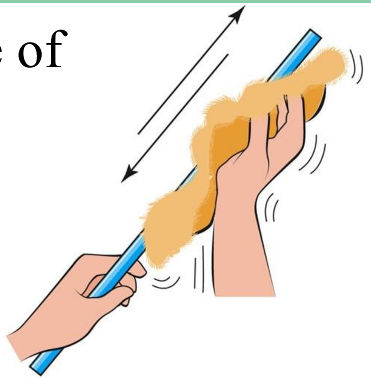


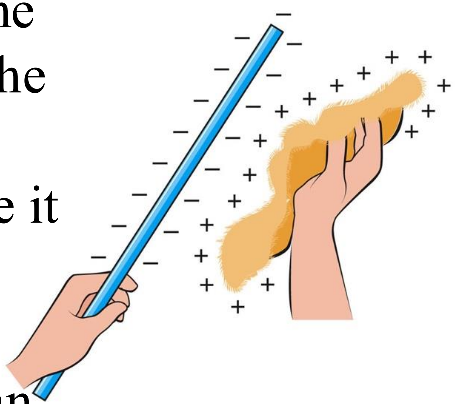
(a) (b)

- For example, if a neutral helium atom gains one electron, it has three negative particles (electrons) and two positive particles (protons). This atom is said to be a negative ion, because it has a net negative charge.
- Similarly, if a neutral helium atom loses one electron, it becomes a positive ion, because it has two positive particles and only one negative particle.

# 7.1 Electric Charge

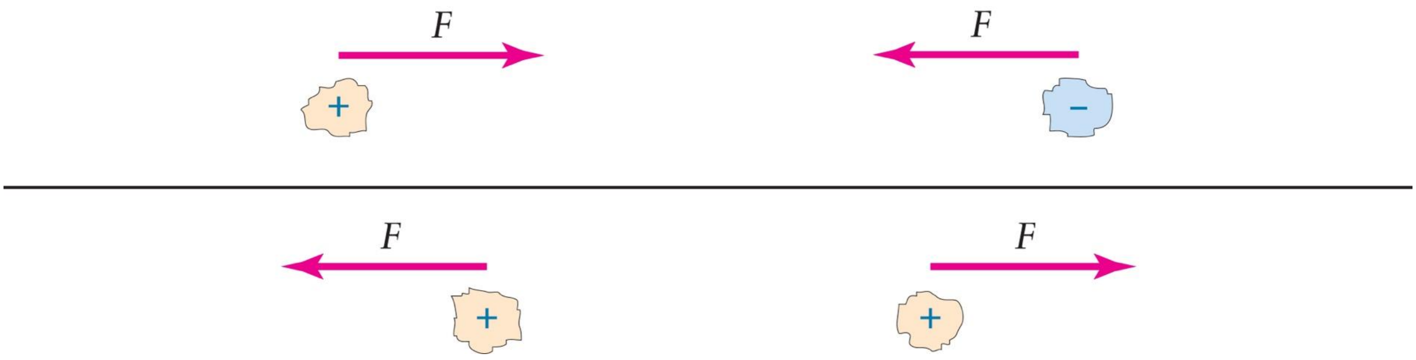
- In many situations, ions are formed on the surface of a substance by the action of friction.
- When a piece of amber, plastic, or hard rubber is rubbed with fur, negative ions are formed on its surface.
  - The contact between the fur and the material causes some of the electrons in the atoms of the fur to be transferred to some of the atoms on the surface of the solid.
  - The fur acquires a net positive charge, because it has fewer electrons than protons.
  - Similarly, the amber, plastic, or hard rubber acquires a net negative charge because it has an excess of electrons.





- Combing your hair can charge the comb in the same way. Rubbing glass with silk causes the glass to acquire a net positive charge.

# 7.2 Electric Force and Coulomb's Law




- You may have noticed hair being drawn toward a charged comb—The negatively charged comb exerts an attractive force on the positively charged hair— two objects with opposite charges attract each other.
- In addition, two objects with the same kind of charge (both positive or both negative) repel each other.

**Like charges repel, unlike charges attract**

# 7.2 Coulomb’s Law

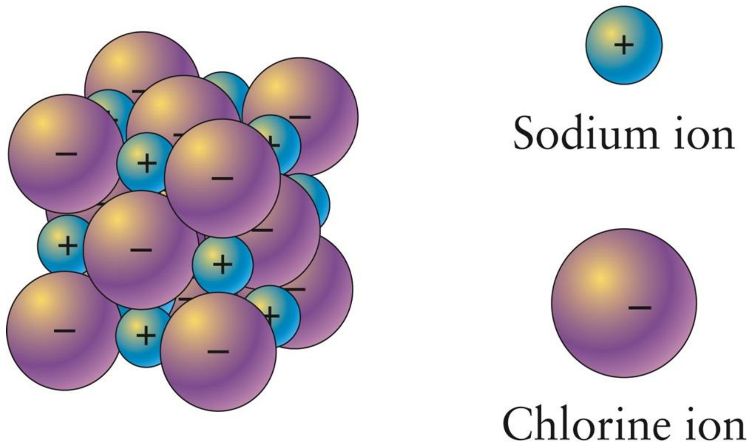
- This force between charged objects is extremely important in the physical world, particularly at the atomic level.

- It is this interaction that holds atoms together and makes it possible for them to exist:

  - In each atom, the positively charged protons in the nucleus exert attractive forces on the negatively charged electrons.

  - The electric force on each electron provides the centripetal force that keeps it in its orbit, much as the gravitational force exerted by the Sun keeps Earth in its orbit.

# 7.2 Coulomb's Law




- For example, when salt is formed from the elements of sodium and chlorine, each sodium atom gives up an electron to a chlorine atom. The resulting ions exert attractive forces on one another because they are oppositely charged.

# 7.2 Coulomb's Law

LAWS

**Coulomb's Law:** The force acting on each of two charged objects is directly proportional to the net charges on the objects and inversely proportional to the square of the distance between them:

$$F \propto \frac{q_1 q_2}{d^2}$$

The constant of proportionality in SI units is:
$9 \times 10^9$ N-m$^2$/C$^2$. Therefore, in SI units:

$$F = \frac{\left(9 \times 10^9\right) q_1 q_2}{d^2}$$

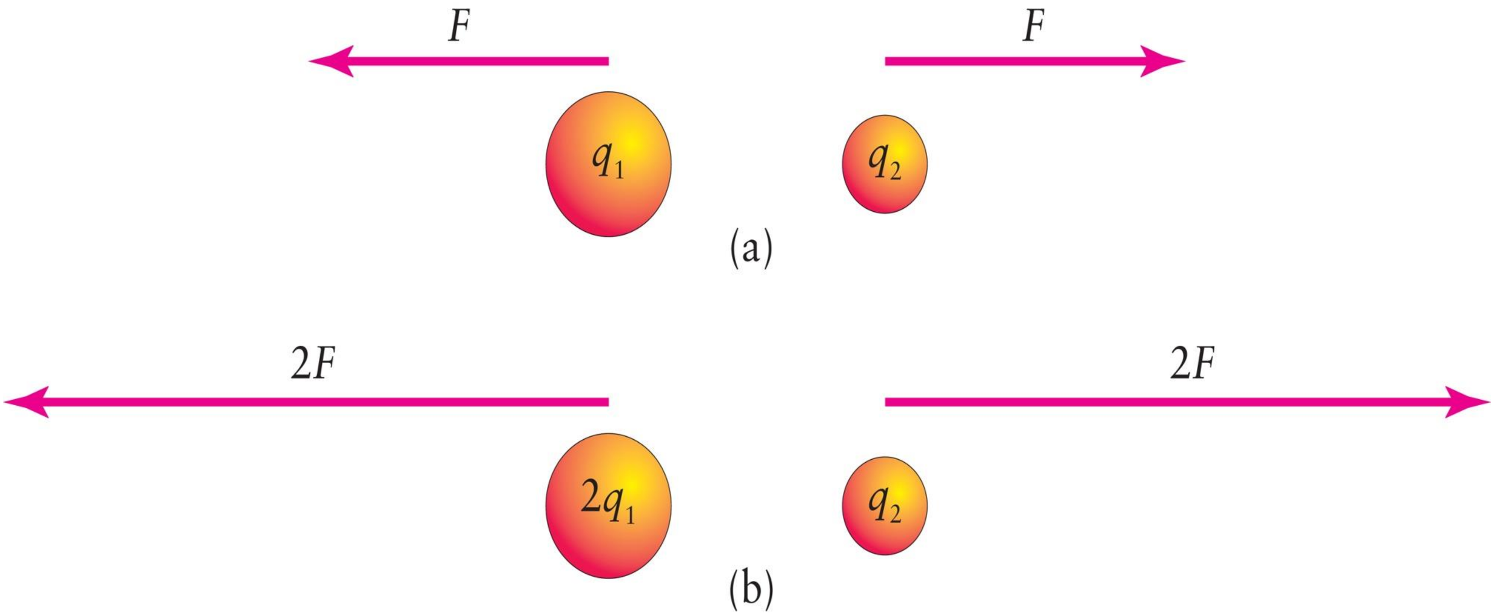
with $F$ in newtons, $q_1$ and $q_2$ in coulombs, and $d$ in meters.

# 7.2 Coulomb's Law

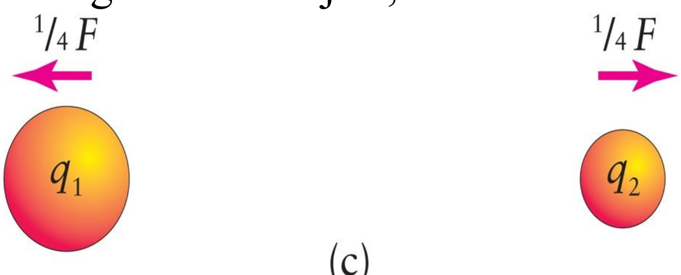
- If two objects have the same kind of charge (both positive or both negative), then the force is *repulsive*.

- If one charge is negative and the other positive, then the force is *attractive*.

# 7.2 Coulomb's Law



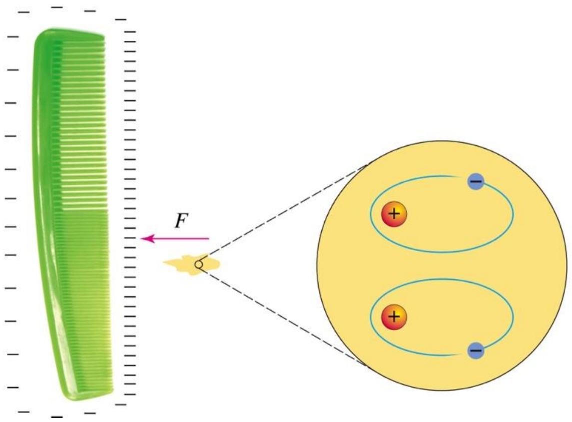

Doubling the charge on one object, doubles the force on both




Doubling the distance between the charges, reduces the force to one-fourth its original value

# 7.2 Coulomb's Law

It is possible for a charged object to exert a force of attraction on a second object that has no net charge.




- ❑ This is what happens when a charged comb is used to pick up bits of paper or thread.
  - Here the negatively charged comb attracts the nuclei of the atoms and repels the electrons.
  - The orbits of the electrons are distorted so that the electrons are, on the average, farther away from the charged comb than the nuclei.
  - This results in a net attractive force because the repulsive force on the slightly more distant, negatively charged electrons is smaller than the attractive force on the closer, positively charged protons.

# 7.2 Coulomb's Law

- The process of inducing a small charge separation (or displacement) between the nucleus of an atom and its electrons is called *polarization*.

- Some molecules are naturally polarized—that is, they have a net negative charge displaced to one side of the net positive nuclear charge. They are called *polar molecules*.

- Water molecules have this property. If a polar molecule is free to rotate—as in a liquid—it will be attracted to a charged object. Its side with the charge opposite that on the object will turn toward the object, and the attractive force on that side will be stronger than the repulsive force on the other side.

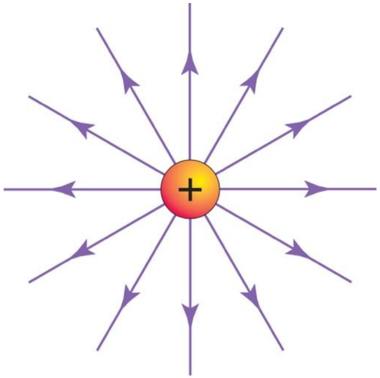
# 7.2 The Electric Field

The electrostatic force is another example of "action at a distance."

- As with gravitation, the concept of a field is useful.
- In the space around any charged object, there is an electric field. This field is the "agent" of the electrostatic force: it will cause any charged object to experience a force.

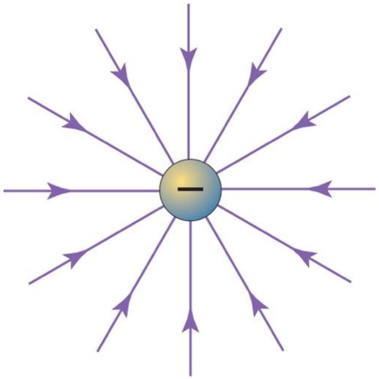
# 7.2 The Electric Field

- The electric field around a charged particle is represented by lines that indicate the direction of the force that the field would exert on a positive charge.



(a)

- (a) The electric field lines around a *positively* charged particle point radially *outward*, and (b) the field lines around a *negatively* charged particle point radially *inward*.



(b)

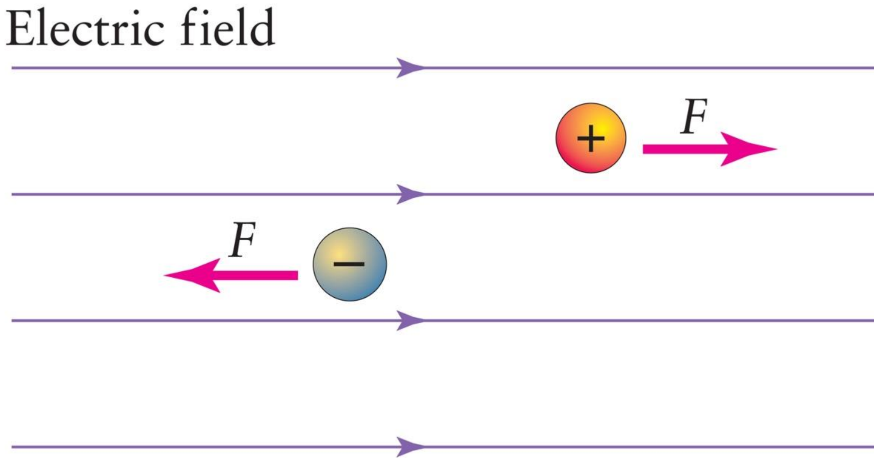
# 7.2 The Electric Field

- The *strength of an electric field* at a point in space is equal to the size of the force that it would cause on a given charged object placed at that point, divided by the size of the charge on the object.

$$\text{electric field strength} = \frac{\text{force on a charged object}}{\text{charge on the object}}$$

- As defined here, the SI units for the electric field are newtons per coulomb: N/C.
- Where the field is strong, a charged object will experience a large force. The electric field around a charged particle is clearly weaker at greater distances from it.
- The strength of the electric field is indicated by the spacing or density of the field lines: where the lines are close together, the field is strong.

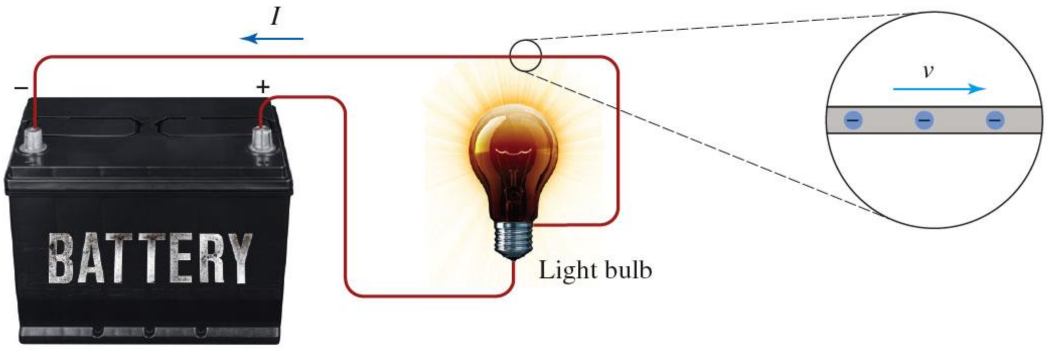
# 7.2 The Electric Field



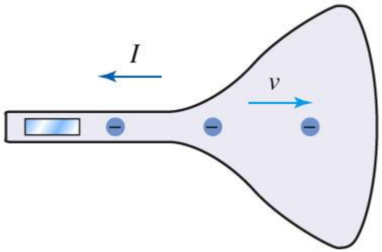

- Any time a *positive* charge is in an electric field, it experiences a force in the *same direction* as the field lines.
- A *negative* charge in an electric field feels a force in the *opposite direction* of the field lines.
- Remember that the electric field exerts the force on a charged object.

# 7.3 Electric Currents—Superconductivity

- An electric current is a flow of charged particles.



(a)



(b)

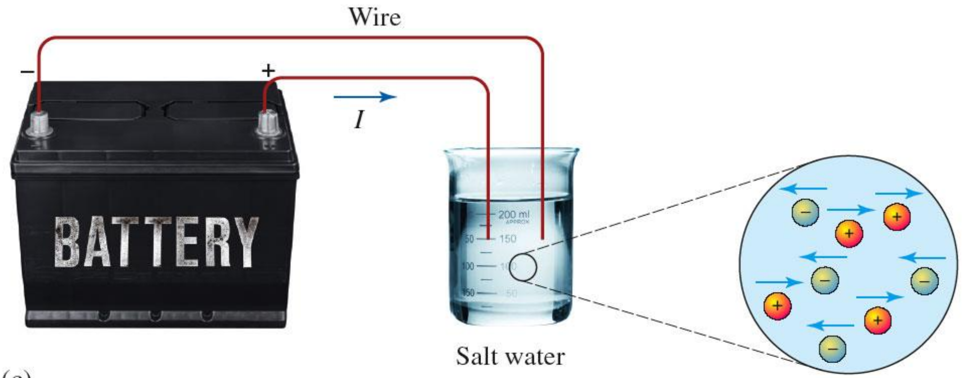
(a) When the appliance is plugged in and operating, electrons flowing inside a metal wire.

(b) There is a near vacuum inside the picture tube, so the electrons can travel without colliding with gas molecules.

# 7.3 Electric Currents—Superconductivity

- An electric current is a flow of charged particles.

(c) When salt is dissolved in water, the sodium and chlorine ions separate and can move about just like the water molecules. If an electric field is applied to the water, the positive sodium ions will flow one way (in the direction of the field), and the negative chlorine ions will flow the other way.



(c)

# 7.3 Electric Current

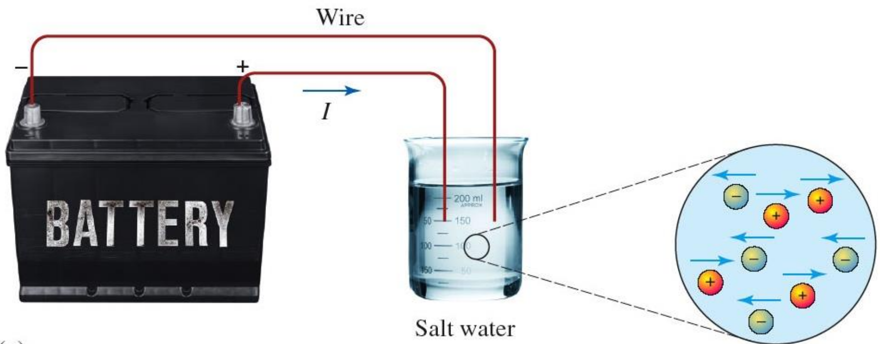
DEFINITION

**Current:** The rate of flow of electric charge. The amount of charge that flows by per second.

$$\text{current} = \frac{\text{charge}}{\text{time}} \qquad I = \frac{q}{t}$$

The SI unit of current is the *ampere* (A or amp), which equals 1 coulomb per second. Current is measured with a device called an *ammeter.*

# 7.3 Electric Current



(c)

- ❑ Either positive charges or negative charges can comprise a current.
- ❑ The effect of a positive charge moving in one direction is the same as that of an equal negative charge moving in the opposite direction.
- ❑ Historically, an electric current is represented as a flow of positive charge. This is because it was originally believed that positive charges moved through metals.
- ❑ Even after it was discovered that it is negatively charged electrons that flow in a wire to comprise the current, the convention of defining the direction of current flow as that which would be associated with positive charges was retained.

- ❑ *If positive ions are flowing to the right in a liquid, then the current is to the right. If negative charges (like electrons) are flowing to the right, then the direction of the current is to the left.*

# 7.3 Electric Current

- Any material that does not readily allow the flow of charges through it is called an electrical *insulator*.
- Substances such as plastic, wood, rubber, air, and pure water are insulators because the electrons are tightly bound in the atoms, and electric fields are usually not strong enough to rip them free so they can move.
- For example: the electricity powering the devices in our homes could kill us if insulators, like the covering on power cords, didn't keep it from entering our bodies.

# 7.3 Electric Current

- An electrical *conductor* is any substance that readily allows charges to flow through it.
- Metals are very good conductors because some of the electrons are only loosely bound to atoms and so are free to “skip along” from one atom to the next when an electric field is present.

# 7.3 Electric Current

- *Semiconductors* are substances that fall in between the two extremes.
- The elements silicon and germanium, both semiconductors, are poor conductors of electricity in their pure states, but they can be modified chemically to have very useful electrical properties.
- Transistors, solar cells, and numerous other electronic components are made out of such semiconductors.

# 7.3 Resistance

DEFINITION

**Resistance:** A measure of the opposition to current flow. Resistance is represented by $R$, and the SI unit of measure is the *ohm* ($\Omega$).

# 7.3 Resistance

- In general, a conductor will have low resistance and an insulator will have high resistance.

- The actual **resistance** of a particular piece of conducting material depends on four factors:

1. **Composition.** The metal making up the wire affects the resistance. For example, an iron wire will have a higher resistance than an identical copper wire.

2. **Length.** The longer the wire is, the higher its resistance.

3. **Diameter.** The thinner the wire is, the higher its resistance.

4. **Temperature.** The higher the temperature of the wire, the higher its resistance.

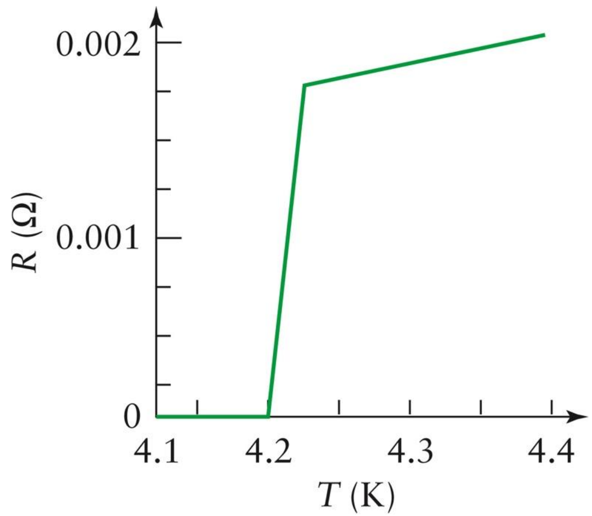
# 7.3 Resistance

- Resistance can be compared to friction.
  - Resistance inhibits the flow of electric charge, and friction inhibits relative motion between two substances.
  - In metals, electrons in a current move among the atoms and in the process collide with them and give them energy. This impedes the movement of the electrons and causes the metal to gain internal energy.

# 7.3 Superconductivity

The larger the current through a particular device, the greater the heating.

- In 1911, Dutch physicist Onnes made an important discovery while measuring the resistance of mercury at extremely low temperatures.

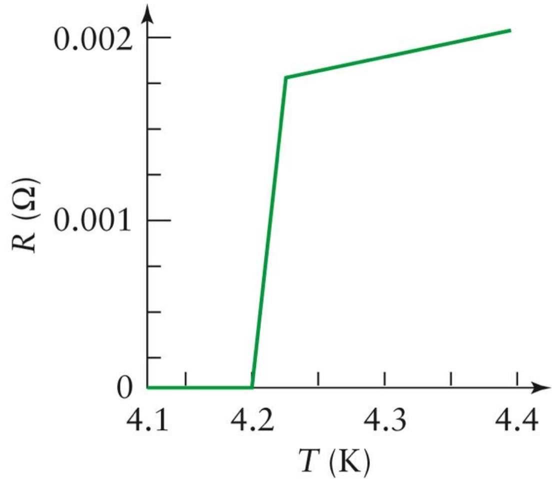



- He found that the resistance decreased steadily as the temperature was lowered, until at 4.2 K (–452.1°F) it suddenly dropped to zero. Electric current flowed through the mercury with no resistance.

# 7.3 Superconductivity

- Onnes named this phenomenon *superconductivity* for good reason: mercury is a perfect conductor of electric current below what is called its *critical temperature* (referred to as $T_c$) of 4.2 K, where electricity flowing through wires with no loss of energy to heating.

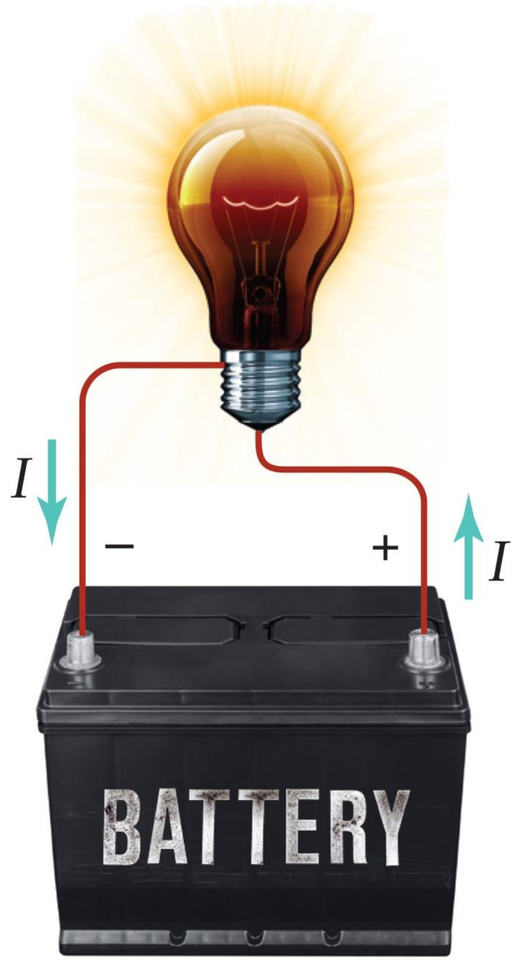
- Subsequent research showed that hundreds of elements, compounds, and metal alloys become superconductors, but only at very low temperatures.



# 7.4 Electric Circuits and Ohm's Law

- An electric circuit is any system consisting of a battery or other electrical power supply, some electrical device such as a lightbulb, and wires or other conductors to carry the current to and from the device.

- The power supply acts like a "charge pump": it forces charges to flow out of one terminal, go through the rest of the circuit, and flow into the other terminal.



# 7.4 Voltage and Ohm's Law

- In a conventional flashlight, for instance, the batteries cause electrons to flow through the bulb's filament.
- Because a force acts on the electrons and causes them to move through a distance, work is done on the electrons by the batteries.
- In other words, the batteries give the electrons energy. This energy is converted into internal energy and light as the electrons go through the lightbulb and heat the filament.
- This leads to the concept of electric voltage.

# 7.4 Voltage and Ohm's Law

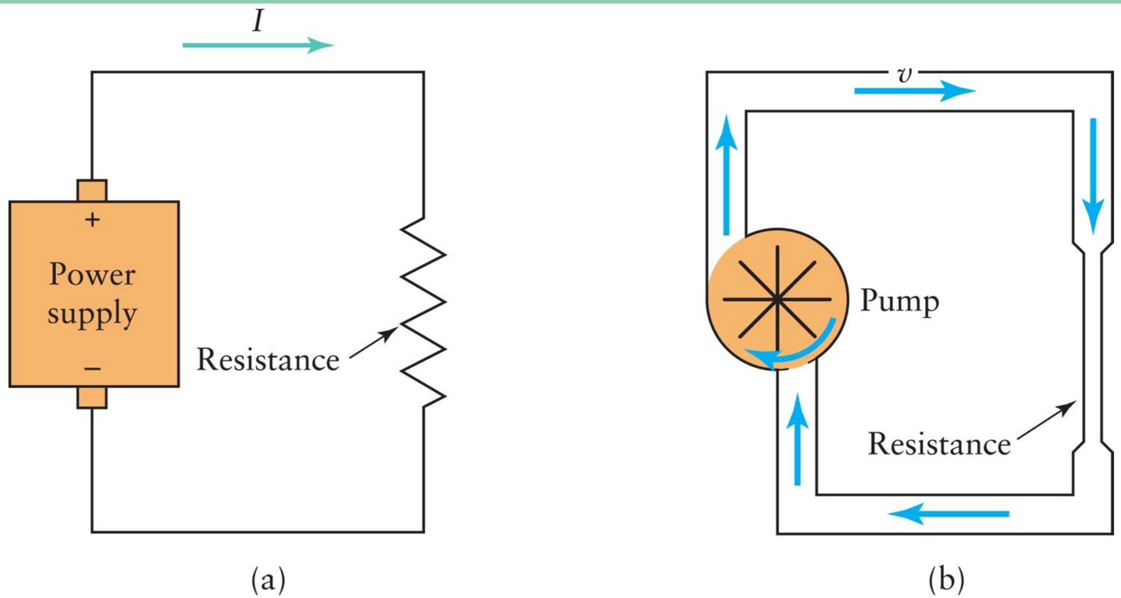
DEFINITION

**Voltage:** The work that a charged particle can do divided by the size of the charge. The energy per unit charge that is provided to charged particles by a power supply.

$$V = \frac{\text{work}}{q} \qquad V = \frac{E}{q}$$

The SI unit of voltage is the *volt* (V), which is equal to *1 joule per coulomb*. Voltage is measured with a device called a *voltmeter*.

# 7.4 Voltage and Ohm's Law




If we return to the analogy of a battery as a charge pump, the voltage plays the role of pressure. A high voltage causing charges to flow in a circuit is like a high pressure causing a fluid to flow.

Even when the circuit is disconnected from the power supply and there is no charge flow, the power supply still has a voltage. In this case, the electric charges have potential energy. Voltage is also referred to as electric potential

# 7.4 Voltage and Ohm's Law

The size of the current that flows through a conductor depends on its resistance and on the voltage causing the current.

LAWS

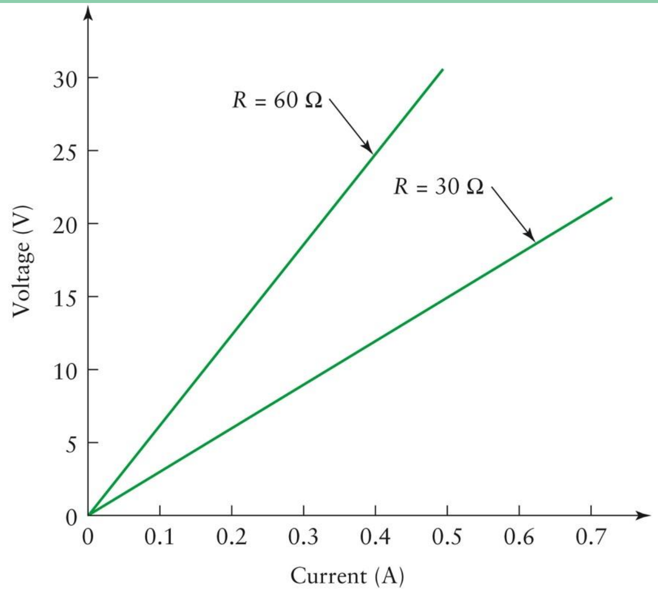
**Ohm's Law**: The current in a conductor is equal to the voltage applied to it divided by its resistance:

$$I = \frac{V}{R} \quad \text{or} \quad V = IR$$

# 7.4 Voltage and Ohm's Law

By Ohm's law: $I = \dfrac{V}{R}$ or $V = IR$

- The higher the voltage for a given resistance, the larger the current.
- The larger the resistance for a given voltage, the smaller the current.




- By applying different-sized voltages to a given conductor, one can produce different-sized currents.
- A graph of the voltage versus the current will be a straight line with a slope that is equal to the conductor's resistance.
- Reversing the polarity of the voltage (switching the "–" and "+" terminals) will cause the current to flow in the opposite direction.

# Example 7.1

A lightbulb used in a 3-volt flashlight has a resistance equal to 6 ohms. What is the current in the bulb when it is switched on?

$$I = \frac{V}{R} = \frac{3\ \text{V}}{6\ \Omega} = 0.5\ \text{A}$$

# Example 7.2

A small electric heater has a resistance of 15 ohms when the current in it is 2 amperes. What voltage is required to produce this current?

$$V = IR = 2\ \mathrm{A} \times 15\ \Omega = 30\ \mathrm{V}$$

# 7.4 Series and Parallel Circuits

- In many situations, several electrical devices are connected to the same electrical power supply. For example, a house may have a hundred different lights and appliances all connected to one cable entering the house. An automobile has dozens of devices connected to its battery.

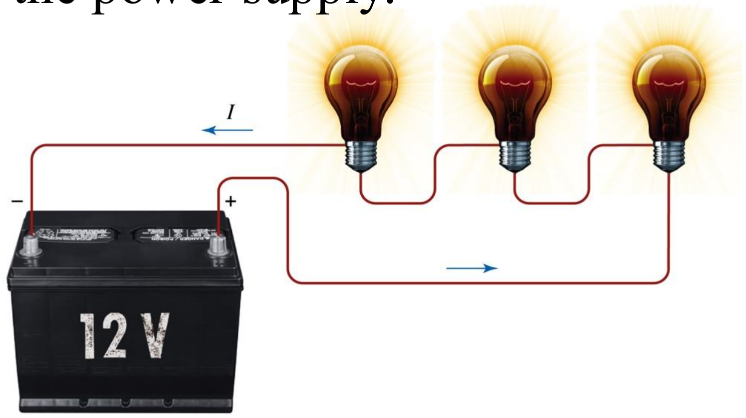
- There are two basic ways in which more than one device can be connected to a single electrical power supply—by a *series circuit* and by a *parallel circuit*.

# 7.4 Series and Parallel Circuits

***In a series circuit***, there is **only one path** for the charges to follow, so **the same current flows in each device**.

- In such a circuit, the ***voltage is divided among the devices***: the voltage on the first device plus the voltage on the second device, and so on, equals the voltage of the power supply.

$$V_{power} = V_1 + V_2 + V_3 + \ldots$$

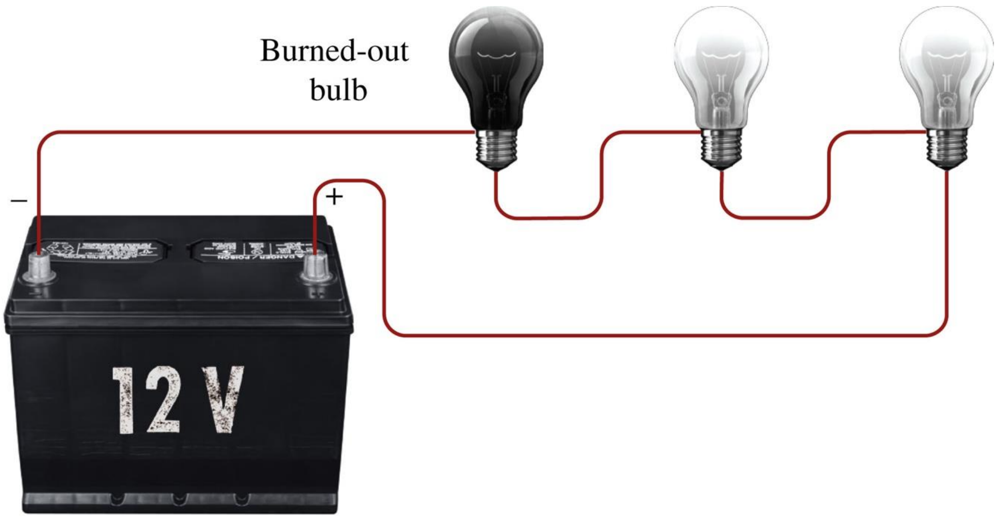



- For example, if three lightbulbs with the same resistance are connected in series to a 12-volt battery, the voltage on each bulb is 4 volts.

# 7.4 Series and Parallel Circuits

***In a series circuit:***

If any of the devices breaks the circuit, the current in a series circuit is stopped. For example, a series circuit with a number of lightbulbs, if one of them burns out, the current stops and all of the bulbs go out.

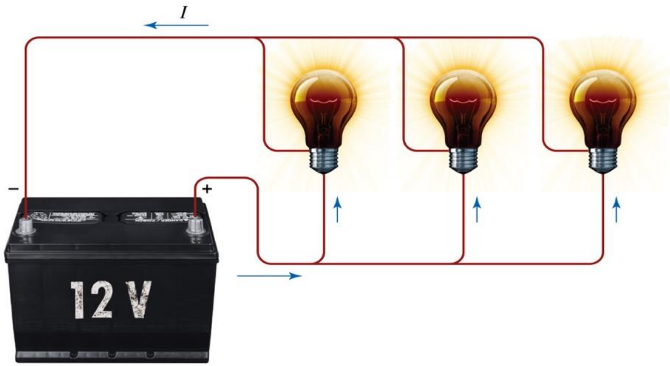


# 7.4 Series and Parallel Circuits

***In a parallel circuit,*** the **current** through the power supply is "**shared**" among the devices while each has the same voltage.

- *The current flowing in the first device plus the current in the second device, and so on, equals the current output by the power supply*.

$$I_{power} = I_1 + I_2 + I_3 + \ldots$$




- There is *more than one path for the current to flow*—in the figure below, three paths. If one of the devices burns out or is removed, the others still function and remain lit.

# Example 7.3

Three lightbulbs are connected in a parallel circuit with a 12-volt battery. The resistance of each bulb is 24 ohms. What is the current produced by the battery?

$$I = \frac{V}{R} = \frac{12\ \text{V}}{24\ \Omega} = 0.5\ \text{A}$$

$$I = 0.5\ \text{A} + 0.5\ \text{A} + 0.5\ \text{A} = 1.5\text{A}$$

# 7.5 Power and Energy in Electric Currents

- Because a battery or other electrical supply must continually put out energy to cause a current to flow, it is important to consider the *power output*—the rate at which energy is delivered to the circuit.

- The power is determined by the voltage of the power supply and the current that is flowing.

- The power supply gives a certain amount of energy to each coulomb of charge that flows through the circuit.

# 7.5 Power and Energy in Electric Currents

- Consequently, *the energy output per unit time* equals the energy given to each coulomb of charge multiplied by the number of coulombs that flow through the circuit per unit time: energy per unit time.

$$\text{energy per unit time} =$$
$$\text{energy per coulomb} \times \text{number of coulombs per unit time}$$

$$\text{power} = \text{voltage} \times \text{current}$$
$$P = VI$$

The units work out correctly in this equation:
joules per coulomb (volts) x coulombs per second (amperes) = joules per second (watts).

# Example 7.4

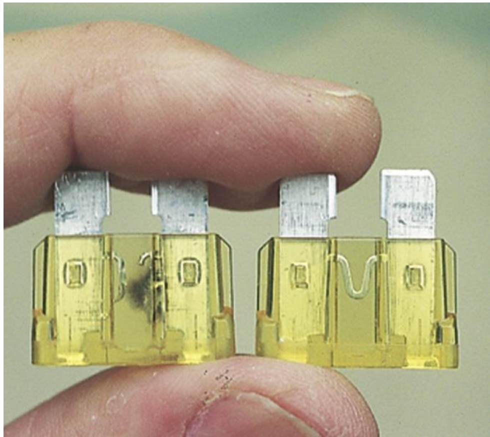
In Example 7.1, we computed the current that flows in a flashlight bulb. What is the power output of the batteries?

$$P = VI = 3 \text{ V} \times 0.5 \text{ A}$$
$$= 1.5 \text{ W}$$

# 7.5 Power and Energy in Electric Currents

A sufficiently large current in any wire can cause it to become very hot—hot enough to melt any insulation around it or to ignite combustible materials nearby.

- *Fuses and circuit breakers* are put into electric circuits as safety devices to *prevent dangerous overheating of wires*.
- If something goes wrong or if too many devices are plugged into the circuit and the current exceeds the recommended safe limit for the size of wire used, the fuse or circuit breaker will automatically "break" the circuit and the current will stop.
- A fuse is a fine wire or piece of metal inside a glass or plastic case. *When the current exceeds the fuse's design limit, the metal melts away, and the circuit is broken*.



# Example 7.5

An electric hair dryer is rated at 1,875 watts when operating on 120 volts. What is the current flowing through it?

$$P = VI$$

$$1{,}875\text{ W} = 120\text{ V} \times I$$

$$\frac{1{,}875\text{ W}}{120\text{ V}} = I$$

$$I = 15.6\text{ A}$$

# 7.5 Power and Energy in Electric Currents

- The electricity delivered to a city, subdivision, or individual house must be transmitted with wires.


- The highest current that can flow in a particular wire without causing excessive heating depends on the size of the wire.

- Because P = VI, using a large voltage makes it possible to transmit the same power with a smaller current.

# 7.5 Power and Energy in Electric Currents

- Customers pay for the electricity supplied to them by electric companies based on the amount of energy they use.
- Recall the equation used to define power:


$$P = E/t \rightarrow E = Pt$$

- A typical household can consume more than a billion joules of electrical energy each month. For this reason, a more appropriately sized unit of measure is employed for ***electrical energy—the kilowatt-hour (kWh).***
- **Energy in kilowatt-hours** is computed by expressing power in kilowatts (kW) and time in hours.

$$1\ \text{kWh} = 1\ \text{kW} \times 1\ \text{h} = 1{,}000\ \text{W} \times 3{,}600\ \text{s} = 3{,}600{,}000\ \text{J}$$

# Example 7.6

If the hair dryer discussed in Example 7.5 is used for 3 minutes, how much energy does it use?

$$t = 3\text{ min} = 3 \times 60\text{ s} = 180\text{ s}$$

$$E = Pt = 1{,}875\text{ W} \times 180\text{ s} = 340{,}000\text{ J}$$

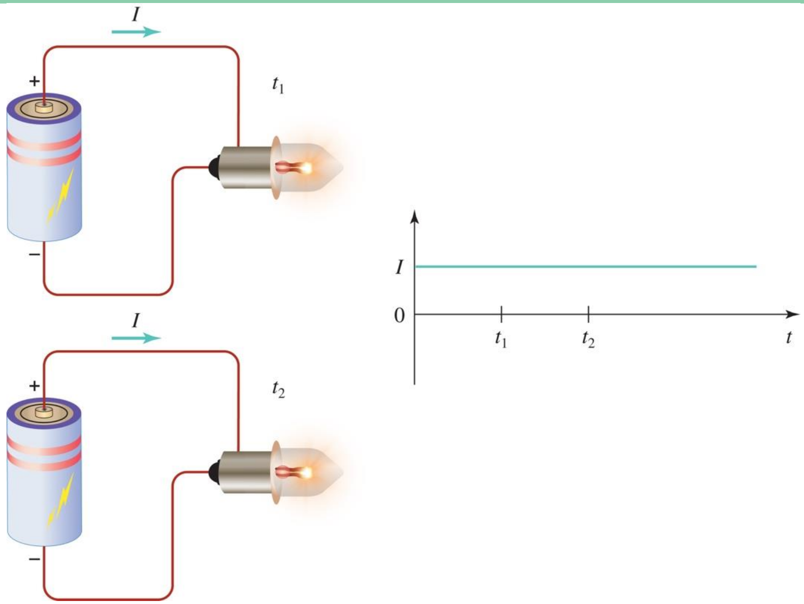
# 7.6 AC and DC

- The electric current supplied by a battery is different from the current supplied by a normal household wall socket.

- Batteries supply ***direct current (DC)***, and household outlets supply ***alternating current (AC)***.

# 7.6 AC and DC

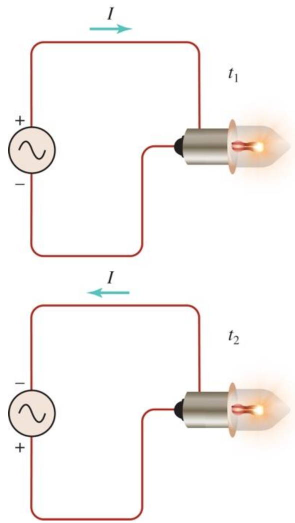
A **DC** power supply, such as a battery, causes a current to flow in a *fixed direction* in a circuit.

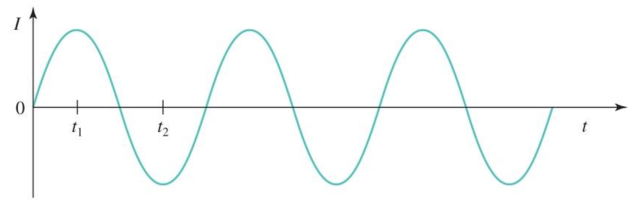



- The current flows out of the positive (+) terminal of the power supply, moves through the circuit, and flows into the negative (–) terminal of the power supply.
- If the total resistance in the circuit doesn’t change, *the size of the current remains constant*.
- A graph of the current versus time is simply a horizontal line.

# 7.6 AC and DC

An **AC** power supply, such as a wall outlet, causes a current to flow in an *alternating direction* in a circuit.




- ❑ In an AC power supply, *the polarity of the two output terminals switches back and forth—the voltage alternates*.
- ❑ This causes the current in any circuit connected to the power supply to alternate as well.
- ❑ Current flows counterclockwise, then clockwise, then back to counterclockwise, and so on.
- ❑ A graph of the current in an AC circuit shows this variation in the size and direction of the current.

# Important Equations

## IMPORTANT EQUATIONS

| Equation | Comments | Equation | Comments |
|---|---|---|---|
| $F = \dfrac{(9 \times 10^9)\, q_1 q_2}{d^2}$ | Coulomb's law (in SI units) | $I = \dfrac{V}{R}$ | Ohm's law |
| $I = \dfrac{q}{t}$ | Definition of current | $V = IR$ | Ohm's law |
| | | $P = VI$ | Electrical power consumption |
| $V = \dfrac{E}{q} = \dfrac{\text{work}}{q}$ | Definition of voltage | $E = Pt$ | Energy consumed during time $t$ |